CELL : THE UNIT OF LIFE

90 MINUTES
SPECIALISED PYQ

Q1. Which of the following functions is carried out by the cytoskeleton in a cell? **{NEET - 2023}**

1.	Transportation	2.	Nuclear division
3.	Protein synthesis	4.	Motility

Q2.Which of the following are NOT considered as the part of endomembrane system? {NEET - 2023}

A. Mitochondria

B. Endoplasmic Reticulum

C. Chloroplasts

D. Golgi complex

E. Peroxisomes

Choose the most appropriate answer from the options given below:

1.	**A, D and E only**	**2.**	**B and D only**
3.	**A, C and E only**	**4.**	**A and D only**

Q3. Given below are two statements: {NEET - 2023}

I:	**In bacteria, the mesosomes are formed by the extensions of the plasma membrane.**
II:	**The mesosomes, in bacteria, help in DNA replication and cell wall formation.**

In light of the above statements, choose the most appropriate answer from the options given below:

1. Statement I is correct but Statement II is incorrect

2. Statement I is incorrect but Statement II is correct

3. Both Statement I and Statement II are correct

4. Both Statement I and Statement II are incorrect

Q4. Which of the following statements are correct in the context of Golgi apparatus?

{NEET - 2023}

A:	It is the important site for the formation of glycoprotein and glycolipids
B:	It produces cellular energy in the form of ATP
C:	It modifies the protein synthesized by ribosomes on ER
D:	It facilitates the transport of ions
E:	It provides mechanical support

Choose the most appropriate answer from the options given below:

1.	(B) and (C) only	2.	(A) and (C) only
3.	(A) and (D) only	4.	(D) and (E) only

Q5.Match List - I with List - II. **{NEET - 2022}**

	List - I		List - II
(a)	Metacentric chromosome	(i)	Centromere situated close to the end forming one extremely short and one very long arms
(b)	Acrocentric chromosome	(ii)	Centromere at the terminal end
(c)	Sub-metacentric	(iii)	Centromere in the middle forming two equal arms of chromosomes
(d)	Telocentric chromosome	(iv)	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(i)	(ii)	(iii)	(iv)
2.	(iii)	(i)	(iv)	(ii)
3.	(i)	(iii)	(ii)	(iv)
4.	(ii)	(iii)	(iv)	(i)

Q6. Given below are two statements: **{NEET - 2022}**

Statement I:	**Mycoplasma can pass through less than 1 micron filter size.**
Statement II:	**Mycoplasma are bacteria with cell walls.**

In the light of the above statements, choose the most appropriate answer from the options given below:
1. Statement I is incorrect but Statement II is correct.
2. Both Statement I and Statement II are correct.
3. Both Statement I and Statement II are incorrect.
4. Statement I is correct but Statement II is incorrect.

Q7. Which of the following statements concerning the Endoplasmic Reticulum is incorrect?
{NEET - 2022}

1. SER are the sites for lipid synthesis
2. RER has ribosomes attached to ER
3. SER is devoid of ribosomes
4. In prokaryotes, only RER is present

Q8. Given below are two statements: **{NEET - 2022}**

Statement I	Membrane-bound organelles of the endomembrane system coordinate cellular functions.
Statement II	Mitochondria and chloroplasts are not considered a part of the endomembrane system.

In the light of the above statements, choose the most appropriate answer from the options given below :

1.	Statement I is correct but Statement II is incorrect.
2.	Statement I is incorrect but Statement II is correct.
3.	Both Statement I and Statement II are correct.
4.	Both Statement I and Statement II are incorrect.

Q9. Match List - I with List - II **{NEET - 2022}**

List - I	List - II
(a) Chromoplasts	(i) Proteins

(b) Amyloplasts	(ii) Oil and fats
(c) Elaioplasts	(iii) Starch
(d) Aleuroplasts	(iv) Carotene

Choose the correct answer from the option given below

	(a)	(b)	(c)	(d)
1.	(iv)	(i)	(iii)	(ii)
2.	(iv)	(iii)	(ii)	(i)
3.	(iv)	(ii)	(iii)	(i)
4.	(iv)	(iii)	(i)	(ii)

Q10.Which type of substances would face difficulty in passing through the cell membrane? **{NEET - 2022}**

1.	**Substances with hydrophobic moiety**
2.	**Substances with hydrophilic moiety**
3.	**All substances, irrespective of hydrophobic and hydrophilic moiety**
4.	**Substances which are soluble in lipids**

Q11. If the pH in lysosomes is increased to alkaline, what will be the outcome? **{NEET - 2022}**

1.	Hydrolytic enzymes will function more efficiently
2.	Hydrolytic enzymes will become inactive

3.	Lysosomal enzymes will be released into the cytoplasm
4.	Lysosomal enzymes will be more active

Q12. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as : **{NEET - 2021}**

1. Sub-metacentric

2. Acrocentric

3. Metacentric

4. Telocentric

Q13. Match List - I with List - II **{NEET - 2021}**

List -I	**List - II**
a. Cristae	**i. Primary constriction in chromosome**
b. Thylakoids	**ii. Disc-shaped sacs in Golgi apparatus**
c. Centromere	**iii. Infoldings in mitochondria**
d. Cisternae	**iv. Flattened membranous sacs in stroma of plastids**

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(iii)	(iv)	(i)	(ii)
2.	(ii)	(iii)	(iv)	(i)
3.	(iv)	(iii)	(ii)	(i)

4.	**(i)**	**(iv)**	**(iii)**	**(ii)**

Q14. Which of the following is an incorrect statement? **{NEET - 2021}**

1.	The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
2.	Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
3.	Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
4.	Microbodies are present both in plant and animal cells.

Q15. The organelles that are included in the endomembrane system are: **{NEET - 2021}**

1.	Golgi complex, Mitochondria, Ribosomes, and Lysosomes
2.	Golgi complex, Endoplasmic reticulum, Mitochondria, and Lysosomes
3.	The endoplasmic reticulum, Mitochondria, Ribosomes, and Lysosomes
4.	The endoplasmic reticulum, Golgi complex, Lysosomes, and Vacuoles

Q16. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells?

{NEET - 2020}

1. Peroxisomes

2. Golgi bodies

3. Polysomes

4. Endoplasmic reticulum

Q17. Which of the following statements about inclusion bodies is incorrect? **{NEET - 2020}**
1. These are involved in the ingestion of food particles.

2. They lie freely in the cytoplasm.

3. These represent reserve material in cytoplasm.

4. They are not bound by any membrane.

Q18. Inclusion bodies of blue-green, purple, and green photosynthetic bacteria are:

{NEET - 2020}

1. Contractile vacuoles

2. Gas vacuoles

3. Centrioles

4. Microtubules

Q19. The biosynthesis of ribosomal RNA occurs in: **{NEET - 2020}**

1. Ribosomes

2. Golgi apparatus

3. Microbodies

4. Nucleolus

Q20. The size of Pleuro Pneumonia Like Organism (PPLO) is: **{NEET - 2020}**

1. 0.02 μm

2. 0.1μm

3. 10-20 μm

4. 0.1μm

Q21. Match the following columns and select the correct option: **{NEET - 2019}**

Column I	Column II

(a)	Smooth endoplasmic reticulum	(i)	Smooth endoplasmic reticulum
(b)	Rough endoplasmic reticulum	(ii)	Lipid synthesis
(c)	Golgi complex	(iii)	Glycosylation
(d)	Centriole	(iv)	Spindle formation

Options:	(a)	(b)	(c)	(d)
1.	(ii)	(i)	(iii)	(iv)
2.	(iii)	(i)	(ii)	(iv)
3.	(iv)	(ii)	(i)	(iii)
4.	(i)	(ii)	(iii)	(iv)

Q22. CThe concept of "*Omnis cellula-e-cellula*" regarding cell division was first proposed by:

{NEET - 2019}

1. Aristotle

2. Rudolf Virchow

3. Theodore Schwann

4. Schleiden

Q23.The shorter and longer arms of submetacentric chromosome are referred to as:

{NEET - 2019}

1. m-arm and n-arm respectively

2. s-arm and l-arm respectively

3. p-arm and q-arm respectively

4. q-arm and p-arm respectively

Q24.Which of the following pair of organelles does not contain DNA? **{NEET - 2019}**

1. Nuclear envelope and Mitochondria

2. Mitochondria and Lysosomes

3. Chloroplast and Vacuoles

4. Lysosomes and vacuoles

Q25. Which of the following statements is not correct? **{NEET - 2019}**

1.	Lysosomes are formed by the process of packaging in the endoplasmic reticulum.
2.	Lysosomes have numerous hydrolytic enzymes
3.	The hydrolytic enzyme of lysosomes are active under acidic pH
4.	Lysosomes are membrane bound structure.

Q26. Which of the following cell organelles is present in the highest number in secretory cells? **{NEET - 2019}**

1.	**Mitochondria**	**2.**	**Golgi complex**
3.	**Endoplasmic reticulum**	**4.**	**Lysosomes**

Q27. Non-membranous nucleoplasmic structures in the nucleus, are the sites for active synthesis of: **{NEET - 2019}**

1.	protein	2.	mRNA

3.	rRNA	4.	tRNA

Q28. Which of the following nucleic acids is present in an organism having 70 S ribosomes only?

{NEET - 2019}

1. Single-stranded DNA with a protein coat
2. Double-stranded circular naked DNA
3. Double-stranded DNA enclosed in nuclear membrane
4. Double-stranded circular DNA with histone proteins

Q29. Match column I with column II. **{NEET - 2019}**

Column I		Column II	
(a)	Golgi apparatus	(i)	Synthesis of protein
(b)	Lysosomes	(ii)	Trap waste and excretory products
(c)	Vacuoles	(iii)	Formation of glycoproteins and glycolipids
(d)	Ribosomes	(iv)	Digesting biomolecules

Choose the right match from the options given below:

Options:	(a)	(b)	(c)	(d)
1.	(iii)	(iv)	(ii)	(i)
2.	(iv)	(iii)	(i)	(ii)
3.	(iii)	(ii)	(iv)	(i)

4.	(i)	(ii)	(iv)	(iii)

Q30. Which of the following is true for nucleolus? **{NEET - 2018}**

1. Larger nucleoli are present in dividing cells.
2. It is a membrane-bound structure.
3. It takes part in spindle formation.
4. It is a site for active ribosomal RNA synthesis.

Q31. The Golgi complex participates in? **{NEET - 2018}**

1. Fatty acid breakdown

2. Formation of secretory vesicles

3. Respiration in bacteria

4. Activation of amino acid

Q32.Which of the following events does not occur in rough endoplasmic reticulum? **{NEET - 2018}**

1. Protein folding
2. Protein glycosylation
3. Cleavage of signal peptide
4. Phospholipid synthesis

Q33.Many ribosomes may associate with a single mRNA to form multiple copies of a polypeptide simultaneously. Such strings of ribosomes are termed as? **{NEET - 2018}**

1. Polysome

2. Polyhedral bodies

3. Plastidome

4. Nucleosome

Q34.Select the incorrect match: **{NEET - 2018}**

	Column I	**Column II**

1.	**Lampbrush**	**Diplotene bivalents chromosomes**
2.	**Allosomes**	**Sex chromosomes**
3.	**Submetacentric**	**L-shaped chromosomes**
4.	**Polytene**	**Oocytes of amphibians chromosomes**

Q35. Which among the following are the smallest living cells, known without a definite cell wall, pathogenic to plants as well as animals and can survive without oxygen?

{NEET - 2017}

(1) *Bacillus*

(2) *Pseudomonas*

(3) *Mycoplasma*

(4) *Nostoc*

Q36.Which of the following components provides sticky character to the bacterial cell?

{NEET - 2017}

1. Nuclear membrane
2. Plasma membrane
3. Glycocalyx
4. Cell Wall

Q37. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP? **{NEET - 2017}**

1. Ribosome
2. Chloroplast
3. Mitochondrion
4. Lysosome

Q38. Select the mismatch. **{NEET - 2016}**

1. Gas vacuoles — Green bacteria Cells

2. Large central vacuoles — Animal cells

3. Protists — Eukaryotes

4. Methanogens — Prokaryotes

Q39.A cell organelle containing hydrolytic enzyme is **{NEET - 2016}**

(1) lysosome

(2) microsome

(3) ribosome

(4) mesosome

Q40. Microtubules are the constituents of **{NEET - 2016}**

(1) spindle fibres, centrioles and cilia

(2) centrioles, spindle fibres and chromatin

(3) centrosome, nucleosome and centrioles

(4) cilia, flagella and peroxisomes

Q41. A complex of ribosomes attached to a single strand of RNA is known as **{NEET - 2016}**

(1) polymer

(2) polypeptide

(3) okazaki fragment

(4) polysome

Q42. Which of the following is not a feature of the plasmids? **{NEET - 2016}**

(a) Circular structure

(b) Transferable

(c) Single-stranded

(d) Independent replication

Q43.Water soluble pigments found in plant cell vacuoles are **{NEET - 2016}**

(1) chlorophylls

(2) carotenoids

(3) anthocyanins

(4) xanthophylls

Q44. Select the wrong statement: **{NEET - 2016}**

1.	Bacterial cell wall is made up of peptidoglycan
2.	Pili and fimbriae are mainly involved in the motility of bacterial cells
3.	Cyanobacteria lack flagellated cells
4.	Mycoplasma is a wall-less microorganism

Q45. Microtubules are the constituents of? **{NEET - 2016}**

1.	spindle fibres, centrioles and cilia
2.	centrioles, spindle fibres and chromatin
3.	centrosome, nucleosome and centrioles
4.	cilia, flagella and peroxisomes

Q46. Mitochondria and chloroplast are: **{NEET - 2016}**

I:	semi-autonomous organelles.
II:	formed by the division of pre-existing organelles and they contain DNA but lack protein-synthesizing machinery.

Q47. Which one of the following options is correct? **{NEET - 2016}**

1. II is true but I is false

2. I is true but II is false

3. Both I and II are false

4. Both I and II are correct

Q48. Which one of the following cell organelles is enclosed by a single membrane? **{NEET - 2016}**

1. Chloroplasts

2. Lysosomes

3. Nuclei

4. Mitochondria

Q49. A complex of ribosomes attached to a single strand of RNA is known as? **{NEET - 2016}**

1. polymer

2. polypeptide

3. okazaki fragment

4. polysome

ANSWER KEY

1.	(4)	11.	(2)	21.	(1)	31.	(2)	41.	(4)
2.	(3)	12.	(3)	22.	(2)	32.	(4)	42.	(3)
3.	(3)	13.	(1)	23.	(3)	33.	(1)	43.	(3)
4.	(2)	14.	(3)	24.	(4)	34.	(4)	44.	(3)
5.	(2)	15.	(4)	25.	(1)	35.	(3)	45.	(2)
6.	(4)	16.	(2)	26.	(2)	36.	(3)	46.	(1)
7.	(4)	17.	(1)	27.	(3)	37.	(3)	47.	(2)
8.	(3)	18.	(2)	28.	(2)	38.	(2)	48.	(2)
9.	(2)	19.	(4)	29.	(1)	39.	(1)	49.	(4)
10.	(2)	20.	(4)	30.	(4)	40.	(1)		